

DOCUMENTATION

1. INTRODUCTION

Security is an important requirement in homes, offices, laboratories, and restricted areas. Traditional lock systems can be vulnerable to key duplication, theft, or loss. To overcome these issues, biometric authentication systems have become increasingly popular. The Fingerprint Based Door Lock System uses fingerprint recognition technology to provide secure access control. Since every individual's fingerprint is unique, only authorized users can unlock the door. The system is designed using an Arduino Uno, fingerprint sensor, servo motor, LEDs, and a buzzer.

2. OBJECTIVE

- To develop a secure door locking system using biometric authentication.
- To eliminate the need for physical keys.
- To provide quick and reliable access to authorized users.
- To enhance security in homes, offices, and institutions.

3. COMPONENTS USED

Components	Quantity
Arduino Uno	1
R305/AS608 Fingerprint Sensor	1
Servo Motor (SG90)	1
Green LED	1
Red LED	1
Buzzer	1
Resistors (220Ω)	2
Jumper Wires	As required
Breadboard	1
Power Supply (5V)	1

4. SYSTEM FLOW

START
|
Place Finger

```

|
Scan Fingerprint
|
Fingerprint Matched?
|
Yes ----- No
|           |
Green LED ON   Red LED ON
Buzzer Beep    Alarm Beep
Unlock Door     Access Denied
|
Lock Door Again
|
END

```

5. WORKING PRINCIPLE

The fingerprint sensor captures the user's fingerprint and converts it into a digital template.

This template is compared with the fingerprints stored in the sensor's database.

When a match is found, the Arduino activates the servo motor to unlock the door.

Simultaneously, the green LED glows and the buzzer emits a confirmation sound.

If no match is found, the Arduino keeps the door locked and activates the red LED along with an alert buzzer sound.

The system continuously monitors fingerprint input and responds accordingly.

6. ADVANTAGES

- High security.
- Easy to use.
- No need for keys.
- Fast authentication process.
- Low maintenance.
- Stores multiple fingerprints.
- Cost-effective solution.

7. APPLICATIONS

- Home Security Systems
- Office Access Control
- Smart Door Locks
- Locker Security
- Laboratories
- Server Rooms
- Attendance and Access Systems

8. RESULT

The Fingerprint Based Door Lock System successfully authenticates registered users and unlocks the door when a valid fingerprint is detected. Unauthorized users are denied access.

The system provides an efficient and reliable biometric security solution.

Circuit diagram

